

Javier Urbina Alarcon

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Google Scholar Profile

Portfolio: <https://javierurbina-bme.github.io>

EDUCATION

- **Technical University of Munich (TUM) – Germany** 04.2026 – 03.2028 (expected)
M.Sc. Biomedical Engineering and Medical Physics
- **PUCP & UPCH – Peru** 03.2019 – 07.2025
B.Sc. Biomedical Engineering
PUCP: Pontificia Universidad Católica del Perú / UPCH: Universidad Peruana Cayetano Heredia
- **Universitat Politècnica de Catalunya – Spain** 02.2022 – 06.2022
Exchange Program in Biomedical Engineering

PROFESSIONAL EXPERIENCE

- **Klinikum rechts der Isar – Technical University of Munich (TUM)** 06.2026 – Present
Student Assistant – Diagnostic and Interventional Neuroradiology (Neuro-Kopf-Zentrum)
 - Support daily clinical operations in a high-volume neuroradiology department equipped with two MRI scanners, one CT scanner, and one angiography suite.
 - Collaborate with multidisciplinary teams of radiologists, neurologists, and intensive care personnel by assisting with the patient preparation, positioning, and workflow coordination.
 - Interact directly with patients, providing guidance, reassurance, and assistance during examinations.
 - Contribute to image acquisition setup, including coil placement and patient positioning for MRI, scan region selection and protocol setup for CT, and support tasks during in vivo angiographic procedures.
 - Apply essential clinical safety practices, including MRI safety levels, radiation protection principles, and sterile workflow awareness in interventional environments.
- **Advanced Simulation Center – UPCH** 01.09.2025 – 31.03.2026
Biomedical Engineer – Operation of medical simulation equipment
 - Operated and provided technical support for high-fidelity patient simulators: NOELLE (Gaumard) and SimMan ALS (Laerdal Medical).
 - Configured and integrated sensors, monitoring systems, and medical devices within clinical simulation environments.
 - Collaborated with medical specialists to validate technical and functional requirements for simulated clinical scenarios.
- **Clínica Internacional – Biomedical Engineering projects department** 02.06.2025 – 14.08.2025
Biomedical Engineering Intern – medical equipment management
 - Assisted in the procurement process of surgical instruments and medical equipment for the expansion of a new hospital tower.
 - Coordinated with physicians to validate technical requirements and instrumentation needs by specialty.
 - Managed and maintained supplier technical documentation (certificates, datasheets, maintenance protocols, service contracts) for 50+ medical devices across 5 specialties.
 - Conducted load testing and recording for the installation of the Hyport R80 ceiling-mounted system.
 - Participated in a virtual training on photon-counting computed tomography (Siemens Naeotom Alpha).

RESEARCH EXPERIENCE

- **Peter S. Allen MRI Research Centre - University of Alberta** 13.01.2025 – 08.05.2025
Research Intern – Supervisor: Dr. Christian Beaulieu
 - Investigated lifespan-related changes in superficial white matter (SWM) using high-resolution diffusion MRI
 - Applied multi-shell Constrained Spherical Deconvolution and Anatomically Constrained Tractography to identify SWM with cortex-adjacent seeding and streamline clustering
 - Processed data from participants and compared diffusion tensor imaging metrics between SWM and deep white matter
 - Conducted exploratory data evaluation and processing of diffusion metrics (e.g., FA, MD) across lifespan stages.
 - Tools used: MRtrix3, Dipy (Python), FreeSurfer, SynthSeg

• Medical Imaging Laboratory - PUCP

01.11.2023 – 06.01.2025

Research Intern – Supervisor: Dr. Andres Coila

- Estimation of nonlinear ultrasound parameters using second harmonic for tissue characterization
- Conducted acoustic field simulations using K-Wave (MATLAB) to model ultrasound propagation, medium attenuation, and sensor responses; developed signal processing pipelines for quantitative tissue characterisation.
- Estimated tissue attenuation maps using regularization techniques to enhance image quality
- Tools used: MATLAB (K-Wave simulation, signal processing)

• Laboratory in Biomechanics and Applied Robotics - PUCP

22.08.2022 – 30.06.2025

Research Intern – Supervisors: Dr. Victoria Abarca and Dr. Dante Elias

- Designed and implemented a real-time therapeutic music-based software interface in C# to improve hand function in children with cerebral palsy by enhancing neuroplasticity
- Processed sensor data from inertial measurement units (IMUs) and flex sensors to track hand and wrist movements
- Conducted data collection sessions with healthy children to assess wrist and finger movement through game-based therapy
- Tools used: Unity (C#), microcontroller programming (C++), MATLAB (signal processing)

• Medical Engineering Laboratory - PUCP

22.08.2022 – 13.05.2023

Research Intern – Supervisor: Dr. Sandra Pérez

- Developed ECG simulation and signal-processing modules for an arrhythmia simulator used in medical device calibration.
- Tools used: microcontroller programming (C++), PCB design, MATLAB (signal processing)

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, C++, C#, R, Unity (game engine), Git/GitHub
- **Medical Imaging:** diffusion MRI tractography, Ultrasound, CT
- **Imaging Software:** MRtrix3, FSL, FreeSurfer, 3D Slicer, ImageJ, SynthSeg, Dipy, K-Wave (MATLAB)
- **Medical Devices:** ECG design & signal processing, embedded systems (C++), patient simulators
- **Clinical & Industry:** Medical device documentation, technical requirements validation, CT/MRI manipulation
- **Standards awareness:** ISO 13485 (documentation context), medical device regulatory frameworks
- **Languages:** Spanish (native), English (C1 – TOEFL iBT 100), German (B1 – actively improving)

AWARDS, SCHOLARSHIPS, AND RESEARCH FUNDING

- **Scholarship:** Emerging Leaders in the Americas Program (ELAP) – Global Affairs Canada **2025**
 - Fully funded scholarship supporting a four-month research internship at the *Peter S. Allen MRI Research Centre*, University of Alberta, Canada.
- **Research Grant:** Annual Research Project Competition (CAP-PUCP) **2022**
 - Agreement No. 2022-C-0008. Funded project: *Soft robotic glove with a music therapy-based playful interface for hand rehabilitation in children with cerebral palsy.*
- **Scholarship:** Exchange Program – Universitat Politècnica de Catalunya (Spain) **2022**
 - Tuition-free exchange program allowing completion of biomedical engineering courses.
- **Research Grant:** BBVA Foundation Peru – COVID-19 Research Funding **2021**
 - Funded project: *Coating of paint based on hydrolyzable tannins and copper applied to handrails in urban transport units to reduce the persistence period of the SARS-CoV-2 virus.*

PUBLICATIONS & PRESENTATIONS

- J. A. Acosta-Franco, **J. Urbina-Alarcón**, A. Madireddy, Z. Sun, C. Weber, P. Smyth, G. Blevins, C. Wilbur, N. Nathoo, D. Emery, C. Beaulieu. High-resolution diffusion tractography shows altered superficial white matter in multiple sclerosis across the lifespan. ISMRM. 2026 (**Oral Presentation**)
- **J. Urbina-Alarcón**, J. A. Acosta-Franco, C. Beaulieu. High-resolution tractography shows later maturation of superficial white matter across the lifespan. International Society of Tractography Conference. 2025 (**Oral Presentation**)
- **J. Urbina-Alarcón**, A. Angulo, V. Abarca, D. Elias. Evaluation of wrist and finger function in healthy children through music-based video game therapy. J Neuroeng Rehabil. 2025;22(1):198 (**Journal Paper**)
- **J. Urbina**, V. Abarca, D. Elias. Integration of music-based game approaches with wearable devices for hand neurorehabilitation: a narrative review. J Neuroeng Rehabil. 2024;21(1):89 (**Journal Paper**)
- R. Muñoz, **J. Urbina**, H. Aleman, J. Rosales, A. Flórez, L. Diaz, M. Córdova, S. Pérez. SonquSim: Design and performance evaluation of an ECG and arrhythmia simulator for medical devices calibration. Annu Int Conf IEEE Eng Med Biol Soc. 2024 (**Poster Presentation**)